The Binomial Theorem

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Binomial coefficients**

Evaluate each binomial coefficient:

a $\binom{6}{2} = 15$ **b** $\binom{8}{0} = 1$ **c** $\binom{5}{5} = 1$ **d** $\binom{10}{3} = 120$

(a) $= 15$; (b) $= 1$; (c) $= 1$; (d) $= 120$

2 **[Foundational]** • **Target: Binomial coefficients**

Write down row 4 of Pascal's triangle (the row beginning $1, 4, \ldots$) and explain how it relates to the expansion of $(a + b)^4$.

Row 4 is $1, 4, 6, 4, 1$ — these are the coefficients of $a^4, a^3b, a^2b^2, ab^3, b^4$ in the expansion.

3 **[Foundational]** • **Target: Expansions and specific terms**

Find the coefficient of x^3 in the expansion of $(2x - 1)^5$.

Term: $\binom{5}{2}(2x)^3(-1)^2 = 10 \cdot 8x^3 = 80x^3$. Coefficient: 80.

4 **[Foundational]** • **Target: Expansions and specific terms**

Use the binomial theorem to expand $(1 + 0.02)^5$ to the x^2-style second-order term, and hence estimate 1.02^5 to four decimal places.

$1 + 5(0.02) + 10(0.02)^2 + \cdots = 1 + 0.1 + 0.004 + \cdots \approx 1.1041$ (true value $1.10408\ldots$).

5 **[Foundational]** • **Target: More expansions and terms**

Find the middle term of the expansion of $(x + 3)^6$ (the term with x^3).

Middle term is the 4th term: $\binom{6}{3}x^3(3)^3 = 20 \cdot 27x^3 = 540x^3$.

6 **[Foundational]** • **Target: More expansions and terms**

Find the constant term in the expansion of $\left(x^2 + \frac{1}{x}\right)^6$.

General term: $\binom{6}{k}(x^2)^{6-k}\left(\frac{1}{x}\right)^k = \binom{6}{k}x^{12-3k}$. Constant term needs $12 - 3k = 0$, so $k = 4$: $\binom{6}{4} = 15$.

7 **[Foundational]** • **Target: Sums of binomial coefficients**

Evaluate $\binom{5}{0} + \binom{5}{1} + \binom{5}{2} + \binom{5}{3} + \binom{5}{4} + \binom{5}{5}$ without computing each term, using the identity $\sum_{k=0}^{n}\binom{n}{k} = 2^n$.

$= 2^5 = 32$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Sums of binomial coefficients**

Use the fact that $\binom{n}{k} = \binom{n}{n-k}$ (symmetry of Pascal's triangle) to explain why $\binom{9}{2} = \binom{9}{7}$ without computing either value directly.

Choosing 2 items to include from 9 is equivalent to choosing 7 items to exclude, so the two counts must be equal — this is the symmetry property $\binom{n}{k} = \binom{n}{n-k}$ with $n = 9$.

9 **[Intermediate]** • **Target: Further expansions**

Find the first three terms in the expansion of $(1 - 2x)^6$ in ascending powers of x.

$\binom{6}{0}1^6 + \binom{6}{1}1^5(-2x) + \binom{6}{2}1^4(-2x)^2 + \cdots = 1 - 12x + 60x^2 - \cdots$.

10 **[Intermediate]** • **Target: Further expansions**

Use the binomial theorem with $(1 - 0.01)^4$ to approximate 0.99^4 to four decimal places, using the first three terms.

$1 + 4(-0.01) + 6(-0.01)^2 + \cdots = 1 - 0.04 + 0.0006 = 0.9606$ (true value 0.96059601).

11 **[Intermediate]** • **Target: Deriving Pascal's triangle recursively**

State Pascal's Rule, $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$, and use it to compute $\binom{6}{3}$ from row 5 of Pascal's triangle ($1, 5, 10, 10, 5, 1$).

$\binom{6}{3} = \binom{5}{2} + \binom{5}{3} = 10 + 10 = 20$.

12 **[Intermediate]** • **Target: The general term formula**

State the general term (the $(k+1)$th term) formula for the expansion of $(a + b)^n$, and use it to find the 5th term of $(x - 2)^8$ directly (without expanding the whole thing).

General term: $T_{k+1} = \binom{n}{k}a^{n-k}b^k$. For the 5th term, $k = 4$: $T_5 = \binom{8}{4}x^4(-2)^4 = 70 \cdot x^4 \cdot 16 = 1120x^4$.

13 **[Intermediate]** • **Target: Combinatorial interpretation of binomial coefficients**

Explain the combinatorial meaning of $\binom{n}{k}$, and use it to explain why $\binom{n}{0} = \binom{n}{n} = 1$ for any n.

$\binom{n}{k}$ counts the number of ways to choose k items from a set of n (order doesn't matter). $\binom{n}{0} = 1$ because there is exactly one way to choose nothing (the empty selection), and $\binom{n}{n} = 1$ because there is exactly one way to choose everything (select the whole set).

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14 [Intermediate] • Target: Applying the binomial theorem to probability-style counting

A fair coin is flipped 5 times. Using $\binom{5}{k}$, find the number of outcomes with exactly 3 heads, and state the total number of possible outcomes (using 2^5).

Outcomes with exactly 3 heads: $\binom{5}{3} = 10$. Total possible outcomes: $2^5 = 32$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: A closing challenge problem • Cross-Lesson: 5.01 Vectors: Components and Operations

Find the sum of all the coefficients in the expansion of $(3x - 2)^4$, using the fact that setting $x = 1$ gives the sum of all coefficients.

Substituting $x = 1$: $(3(1) - 2)^4 = (1)^4 = 1$. The sum of all coefficients in the expansion is 1.

16 [Challenging] • Target: A closing challenge problem • Cross-Lesson: 5.02 The Dot Product

Find the coefficient of x^5 in the expansion of $(x^2 + 3)(x - 1)^6$, by finding the relevant terms from $(x - 1)^6$ first.

From $(x - 1)^6$, the terms of degree 3 and 5 are needed (since multiplying by x^2 or by the constant 3 must produce x^5). Term with x^3: $\binom{6}{3}x^3(-1)^3 = -20x^3$; multiplying by x^2 gives $-20x^5$. Term with x^5: $\binom{6}{1}x^5(-1)^1 = -6x^5$; multiplying by 3 gives $-18x^5$. Total coefficient: $-20 + (-18) = -38$.

17 [Challenging] • Target: A closing challenge problem • Cross-Lesson: 5.03 Parametric Equations

Use the binomial theorem to expand $(x + h)^3 - x^3$ fully, then divide by h and simplify — this expression is exactly the difference quotient used to find the derivative of $f(x) = x^3$ in calculus.

$(x + h)^3 = x^3 + 3x^2h + 3xh^2 + h^3$ (binomial expansion). So $(x + h)^3 - x^3 = 3x^2h + 3xh^2 + h^3$. Dividing by h: $\frac{3x^2h + 3xh^2 + h^3}{h} = 3x^2 + 3xh + h^2$ — as $h \to 0$, this approaches $3x^2$, the derivative of x^3.

18 [Challenging] • Target: A closing challenge problem • Cross-Lesson: 6.01 Polar Coordinates and Conversions

Evaluate $\binom{7}{2}$ and $\binom{7}{5}$, and confirm they are equal using the symmetry property.

$\binom{7}{2} = 21$ and $\binom{7}{5} = 21$. Equal, confirming $\binom{n}{k} = \binom{n}{n-k}$ with $n = 7$, $k = 2$.

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19 [Challenging] • Target: Applying the binomial theorem to probability-style counting • Cross-Lesson: 6.02 Polar Equations and Graphs

A school committee of 4 members is to be selected from a club consisting of 6 juniors and 5 seniors. (a) In how many total ways can the 4 members be selected? (b) In how many ways can the committee be formed if it must contain exactly 2 juniors and 2 seniors?

(a) Unrestricted selection: Total students $= 6 + 5 = 11$. Choosing 4 where order does not matter: $\binom{11}{4} = \frac{11 \times 10 \times 9 \times 8}{4 \times 3 \times 2 \times 1} = 11 \times 10 \times 3 = \mathbf{330}$ ways. (b) Exactly 2 juniors and 2 seniors: Select 2 juniors from 6, AND 2 seniors from 5: $\binom{6}{2} \times \binom{5}{2} = \binom{6 \times 5}{2} \times \binom{5 \times 4}{2} = 15 \times 10 = \mathbf{150}$ ways.

20 [Challenging] • Target: A closing challenge problem • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

A game pays $x15ifaplayerrollsa6onafairsix – sideddie, pays\6$ if they roll a 5, and the player must pay $x3$ for any other roll (1, 2, 3, or 4). Determine the expected gain or loss per roll, and state whether the game is favorable to the player.

Step 1 (Probability Breakdown): $P(\text{roll } 6) = 1/6$ with gain $x15$. P(\text{roll } 5) = 1/6$withgain$+\6. $P(\text{roll } 1, 2, 3, \text{ or } 4) = 4/6$ with loss $x3$. $Step2(ExpectedValue)$: E(X) = 15\left(\frac{1}{6}\right) + 6\left(\frac{1}{6}\right) + (-3)\left(\frac{4}{6}\right) = \frac{15 + 6 - 12}{6} = \frac{9}{6} = \mathbf{+\1.50 Since $E(X) > 0$, the game is favorable to the player with an average expected profit of $x1.50} per roll.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Binomial coefficients

Three distinct vertices of a regular 12-gon are selected uniformly at random. What is the probability that the triangle formed by these three vertices is obtuse?

a. $\frac{3}{11}$

b. $\frac{5}{11}$

c. $\frac{6}{11}$ ✓

d. $\frac{7}{11}$

To find the probability that a randomly chosen triangle is obtuse, we determine the total number of possible triangles and the number of obtuse triangles. **Step 1: Total number of triangles** Choosing any 3 distinct vertices from the 12 vertices of the regular 12-gon gives: $\binom{12}{3} = \frac{12 \times 11 \times 10}{3 \times 2 \times 1} = 220$ **Step 2: Counting right and acute triangles** - **Right Triangles:** By Thales' Theorem, an inscribed triangle is a right triangle if and only if one of its sides is a.

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22 [Foundational] • Target: Expansions and specific terms

The first three terms of the binomial expansion of $(1 + kx)^n$ are $1 + 6x + 12x^2$. Find the value of k.

a. **2** ✓

b. -2

c. 3

d. -3

The binomial expansion of $(1 + x)^n$ is given by $1 + nx + \frac{n(n-1)}{2!}x^2 + \ldots$. For the given expansion $(1 + kx)^n$, we can substitute $y = kx$ to get $(1 + y)^n$. The first three terms of the expansion are $1 + ny + \frac{n(n-1)}{2!}y^2 + \ldots$. Substituting $y = kx$ back, we have: $1 + n(kx) + \frac{n(n-1)}{2!}(kx)^2 + \ldots$ $1 + nkx + \frac{n(n-1)}{2!}k^2x^2 + \ldots$ We are given that the first three terms are $1 + 6x + 12x^2$. Comparing the coefficients: 1. Coefficient of x: $nk = 6$ 2. Coefficient of.

23 [Foundational] • Target: More expansions and terms

A geometric series has first term $a = 11$ and common ratio $r = 1/2$. Find S_∞.

a. 11/2

b. **22** ✓

c. 22/3

d. 33

Since $|r| < 1$, $S_\infty = \frac{a}{1-r} = \frac{11}{1-(1/2)} = 22$

24 [Foundational] • Target: Sums of binomial coefficients

A fair coin is tossed 3 times. Let X be the number of heads obtained. What is the probability that X = 2?

a. $\frac{1}{8}$

b. $\frac{3}{8}$ ✓

c. $\frac{1}{2}$

d. $\frac{7}{8}$

This is a binomial distribution scenario. The number of trials is $n = 3$ (the number of coin tosses). The probability of success (getting a head) in a single trial is $p = \frac{1}{2}$, and the probability of failure (getting a tail) is $q = 1 - p = \frac{1}{2}$. We want to find the probability of getting exactly $k = 2$ heads. The binomial probability formula is $P(X = k) = \binom{n}{k}p^k q^{n-k}$. Substituting the values: $P(X = 2) = \binom{3}{2}(\frac{1}{2})^2(\frac{1}{2})^{3-2}$.

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25 [Foundational] • Target: Further expansions

The bar graph below shows the distribution of customer ratings given to a local bakery by 20 customers on a scale from 1 to 5 stars. How many additional 5-star ratings must the bakery receive so that the overall mean rating of all customers becomes exactly 3.5 stars?

a. 6

b. **8** ✓

c. 12

d. 16

First, calculate the current sum of the ratings from the bar graph:

$\text{Sum} = (1 \times 2) + (2 \times 6) + (3 \times 6) + (4 \times 4) + (5 \times 2) = 2 + 12 + 18 + 16 + 10 = 58$ The current number of customers is $2 + 6 + 6 + 4 + 2 = 20$. Let x represent the number of additional 5-star ratings. Adding x ratings of 5 stars increases the total sum of ratings by $5x$ and increases the total number of customers to $20 + x$. Set up the equation for the new mean: $\frac{58 + 5x}{20 + x} = 3.5$ Multiply both.

26 [Foundational] • Target: Deriving Pascal's triangle recursively

The bar graph below shows the number of students enrolled in each of four elective classes at a high school. Each student is enrolled in exactly one of these four classes. If one student is selected at random from these four classes, what is the probability that the student is enrolled in Drama?

a. $\frac{1}{4}$

b. $\frac{4}{25}$ ✓

c. $\frac{4}{21}$

d. $\frac{7}{25}$

First, find the total number of students enrolled in the four classes by adding the frequencies from the bar graph: $\text{Total} = 12 + 16 + 8 + 14 = 50$ From the graph, the number of students enrolled in Drama is 8. The probability of selecting a student enrolled in Drama is the ratio of the number of Drama students to the total number of students: $P(\text{Drama}) = \frac{8}{50} = \frac{4}{25}$ - $\frac{1}{4}$ incorrectly assumes each elective has an equal number of students. - $\frac{4}{21}$ incorrectly d.

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27 [Foundational] • Target: The general term formula

A fair six-sided die is rolled 4 times. Let X be the number of times a 6 is obtained. The possible values for X are 0, 1, 2, 3, and 4. The probability of obtaining a 6 on any given roll is $p = \frac{1}{6}$. The number of trials is $n = 4$. This scenario can be modelled by a binomial distribution, denoted as $X \sim B(n, p)$. Which of the following represents the correct parameters for this binomial distribution?

a. $B(4, 1/6)$ ✓

b. $B(6, 1/4)$

c. $B(1, 4/6)$

d. $B(4, 4/6)$

The problem describes a situation with a fixed number of independent trials (rolling a die 4 times), where each trial has only two possible outcomes (obtaining a 6 or not obtaining a 6), and the probability of success (obtaining a 6) is constant for each trial. In a binomial distribution, denoted as $X \sim B(n, p)$, n represents the number of trials, and p represents the probability of success on a single trial. From the question: - The number of times the die is rolled is 4, so $n = 4$. - The probability of o.

28 [Foundational] • Target: Combinatorial interpretation of binomial coefficients

The bar chart below displays the number of cars sold by an auto dealership each day from Monday through Thursday. What was the mean number of cars sold per day over these 4 days?

a. 5.5

b. **6.0** ✓

c. 6.5

d. 8.0

To find the mean number of cars sold per day, sum the number of cars sold across all 4 days and divide by the total number of days (4): $\text{Total cars sold} = 5 + 9 + 4 + 6 = 24$ $\text{Mean} = \frac{24}{4} = 6.0$ - The distractor 5.5 is the median of the data set (ordered: 4, 5, 6, 9; median = $\frac{5+6}{2} = 5.5$). - The distractor 6.5 is the midrange ($\frac{4+9}{2} = 6.5$). - The distractor 8.0 results from incorrectly dividing the sum by 3 instead of 4.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Applying the binomial theorem to probability-style counting

State and prove the Bayes' theorem for continuous distributions.

a. Not valid for continuous distributions

b. Same as discrete case

c. $f(\theta|x) = \frac{f(x|\theta)\pi(\theta)}{\int f(x|\theta)\pi(\theta)d\theta}$ **— from the definition of conditional density ✓**

d. Use the CLT

Applying Probability & Statistics principles, Continuous Bayes' theorem. Therefore, the verified correct option is $f(\theta|x) = \frac{f(x|\theta)\pi(\theta)}{\int f(x|\theta)\pi(\theta)d\theta}$ — from the definition of conditional density.

30 [Intermediate] • Target: A closing challenge problem

Show that the binomial distribution approaches Poisson as $n \to \infty$, $p \to 0$, $np = \lambda$ fixed.

a. Requires $n > 100$

b. Not true

c. $\binom{n}{k}p^k(1-p)^{n-k} \to \frac{e^{-\lambda}\lambda^k}{k!}$ **— use** $(1-\lambda/n)^n \to e^{-\lambda}$ **✓**

d. Only for $k = 0$

Applying Probability & Statistics principles, Poisson limit theorem. Therefore, the verified correct option is $\binom{n}{k}p^k(1-p)^{n-k} \to \frac{e^{-\lambda}\lambda^k}{k!}$ — use $(1-\lambda/n)^n \to e^{-\lambda}$.

31 [Intermediate] • Target: Binomial coefficients

State and apply the Central Limit Theorem.

a. Only for discrete distributions

b. **For large** n**,** $\bar{X} \sim N(\mu, \sigma^2/n)$ **approximately, regardless of the population distribution ✓**

c. Only for $n > 1000$

d. Only for normal populations

Applying Probability & Statistics principles, CLT statement. Therefore, the verified correct option is For large n, $\bar{X} \sim N(\mu, \sigma^2/n)$ approximately, regardless of the population distribution.

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32 [Intermediate] • Target: Expansions and specific terms

A factory tests 30 independent components, each with a fixed probability of being defective; X is the number of defective components. Is X binomially distributed?

a. Yes — fixed, independent trials with constant success probability ✓

b. No — the number of trials is not fixed

c. No — trials are not independent / probability is not constant

d. Yes — but only if $n > 30$

The trials are fixed in number, independent, and each has the same probability of success, so the conditions for a binomial distribution are satisfied.

33 [Intermediate] • Target: More expansions and terms

State Bayes' theorem: $P(A|B) =$

a. $\frac{P(B|A)P(A)}{P(B)}$ ✓

b. $\frac{P(A)}{P(B)}$

c. $\frac{P(A)P(B)}{P(A|B)}$

d. $P(A)P(B|A)$

Applying Probability & Statistics principles, Bayes' theorem. Therefore, the verified correct option is $\frac{P(B|A)P(A)}{P(B)}$.

34 [Intermediate] • Target: Sums of binomial coefficients

Of 27 students, 17 play soccer and 3 of those also play basketball. What is the probability that a randomly chosen soccer player also plays basketball?

a. $\frac{3}{10}$

b. $\frac{14}{17}$

c. $\frac{3}{17}$ ✓

d. $\frac{17}{27}$

Conditional probability restricts to the soccer group: $P(\text{basketball} \mid \text{soccer}) = \frac{3}{17} = \frac{3}{17}$.

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35 [Intermediate] • Target: Further expansions

A machine's failure probability increases each hour it runs; X is the number of failures in a day. Is X binomially distributed?

a. No — trials are not independent / probability is not constant ✓

b. Yes — fixed, independent trials with constant success probability

c. No — the number of trials is not fixed

d. Yes — but only if $n > 30$

The conditions for a binomial distribution (fixed independent trials with constant probability) are not satisfied here.

36 [Intermediate] • Target: Deriving Pascal's triangle recursively

The Fundamental Theorem of Algebra states that every polynomial of degree $n \geq 1$ with complex coefficients:

a. Has n rational roots

b. Has at most n roots

c. Has real roots only

d. Has exactly n roots in $\mathbb{C}$ (counting multiplicity) ✓

FTA: every non-constant polynomial has exactly n complex roots (with multiplicity).

37 [Intermediate] • Target: The general term formula

Use De Moivre's theorem to express $\cos 4\theta$ in terms of $\cos\theta$.

a. $\cos^4\theta$

b. $4\cos^2\theta - 3$

c. $8\cos^4\theta - 8\cos^2\theta + 1$ ✓

d. $4\cos^4\theta - 3\cos^2\theta$

Expand $(\cos\theta + i\sin\theta)^4$ and take real part.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Combinatorial interpretation of binomial coefficients • Cross-Lesson: 6.01 Polar Coordinates and Conversions

Derive the maximum likelihood estimator for p in Binomial distribution.

a. $\hat{p} = \sqrt{k/n}$

b. $\hat{p} = k/n$ **where** k **is observed successes and** n **is total trials** ✓

c. $\hat{p} = 1$

d. $\hat{p} = n/k$

Log-likelihood differentiated with respect to p.

39 [Challenging] • Target: Applying the binomial theorem to probability-style counting • Cross-Lesson: 6.02 Polar Equations and Graphs

A bag has 13 marbles, 2 of which are red. What is the probability of drawing a red marble?

a. $\frac{11}{13}$

b. $\frac{2}{13}$ ✓

c. $\frac{1}{13}$

d. $\frac{2}{11}$

$P(\text{red}) = \frac{2}{13} = \frac{2}{13}$.

40 [Challenging] • Target: A closing challenge problem • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

How many ways can you choose 5 items from 13 distinct items, where order does not matter?

a. 1286

b. 1288

c. 154440

d. **1287** ✓

$C(13, 5) = \frac{13!}{5!\,(13-5)!} = 1287$.

41 [Challenging] • Target: Binomial coefficients • Cross-Lesson: 7.02 Logarithms and Their Properties

How many ways can you arrange 2 items chosen from 10 distinct items, where order matters?

a. 89

b. **90** ✓

c. 91

d. 45

$P(10, 2) = \frac{10!}{(10-2)!} = 90$.

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42 [Challenging] • Target: Expansions and specific terms • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

The Poisson distribution arises as a limit of Binomial. For what parameters?

a. $n \to \infty$, $p = 1/2$

b. $n = 1$, any p

c. $p \to 1$

d. $n \to \infty$, $p \to 0$, $np = \lambda$ **constant** ✓

Poisson limit theorem conditions.

43 [Challenging] • Target: More expansions and terms • Cross-Lesson: 7.04 Applications: Growth and Decay

The diagram shows a triangle ABC with coordinates A(2, 1), B(8, 1) and C(5, 5). Find the length of the side AC.

a. 5

b. x ✓

c. 6

d. 7

The coordinates of A are (2, 1) and the coordinates of C are (5, 5). Using the distance formula, $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. Length of AC = $\sqrt{(5-2)^2 + (5-1)^2}$ Length of AC = $\sqrt{(3)^2 + (4)^2}$ Length of AC = $\sqrt{9 + 16}$ Length of AC = $\sqrt{25}$ Length of AC = 5.

44 [Challenging] • Target: Sums of binomial coefficients • Cross-Lesson: 8.01 Rational Functions and Asymptotes

A solid is formed by a cylinder of radius 3 and height 13 topped with a hemisphere of the same radius. Find the total volume. (In terms of π.)

a. 164π

b. 27π

c. 135π ✓

d. 51π

Cylinder $= \pi r^2 h = 117\pi$. Hemisphere $= \frac{2}{3}\pi r^3 = 18\pi$. Total $= 135\pi$.

45 [Challenging] • Target: Further expansions • Cross-Lesson: 8.02 Graphing Rational Functions

Find the volume of a cone with radius 2 and height 15. (In terms of π.)

a. 4π

b. 2π

c. 32π

d. 20π ✓

$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(2)^2(15) = 20\pi$.

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